Black & White Cats

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Agenda

1. Introduction
2. Problem Statement
3. Proposed Solution
4. Key Features
5. System Architecture
6. Alerting and Reporting
7. Testing and Validation
8. Applications and Future Scope



LINUX STORAGE MONITORING SYSTEM

Problem Statement

Administrators need early warnings instead of discovering storage failures after they occur.

Solution

What Black&WhiteCats does

Black&WhiteCats is a storage monitoring system that collects operating-system data and displays actionable storage information.

- Monitors mounted filesystems.
- Tracks disk utilization and inode usage.
- Analyses storage growth and estimates time until full.
- Generates alerts and daily reports.
- Displays information on a live web dashboard.

Key Features

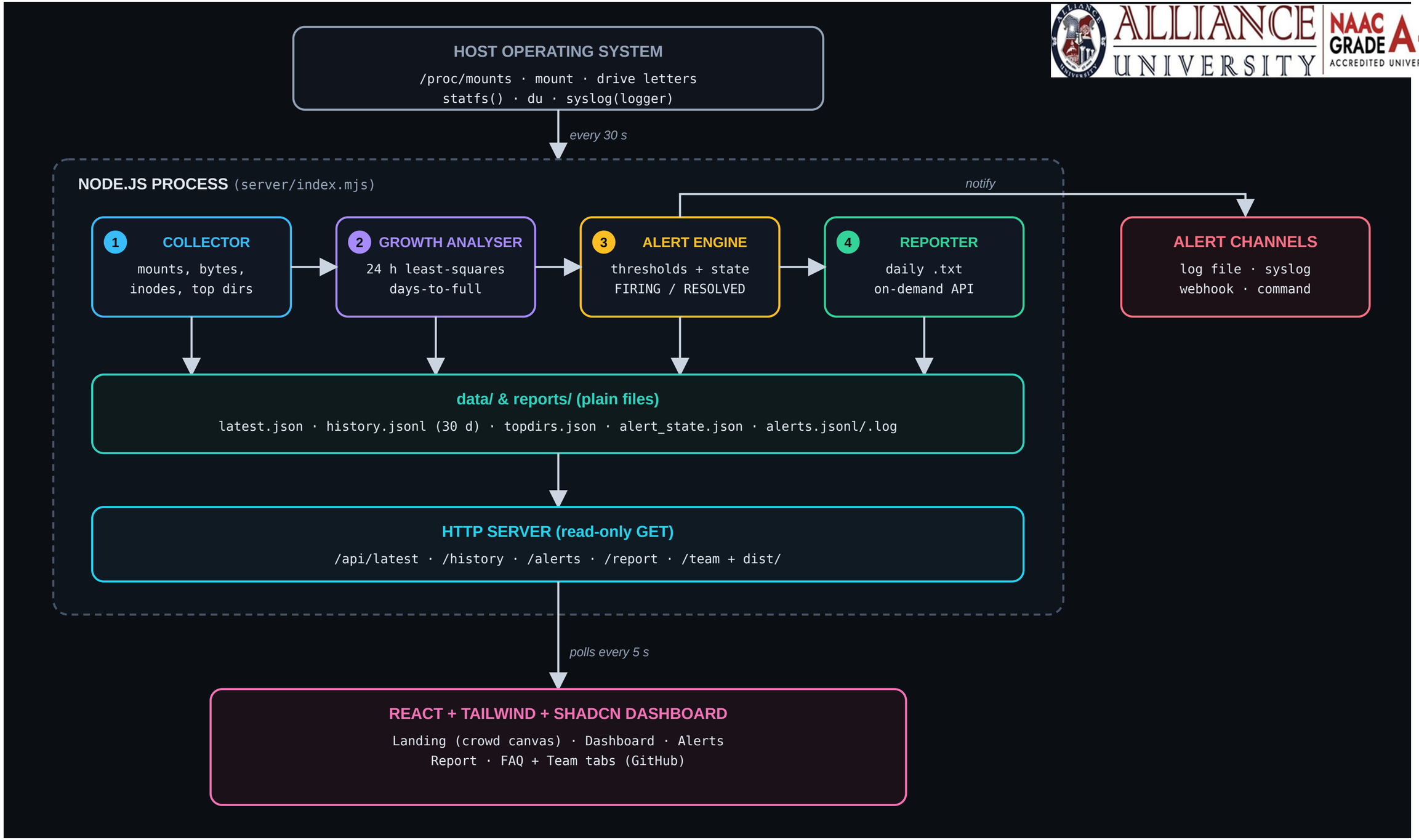
Main capabilities

- Real-time monitoring: Samples storage every 30 seconds.
- Inode monitoring: Detects file-system resource exhaustion.
- Growth analysis: Estimates daily growth and days until full.
- Smart alerts: Tracks warning, critical, and resolved states.
- Daily reports: Generates text-based storage reports.
- Cross-platform support: Linux, macOS, and Windows.

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SYSTEM ARCHITECTURE




HOST OPERATING SYSTEM
/proc/mounts · mount · drive letters
statfs() · du · syslog(logger)
every 30 s
NODE.JS PROCESS (server/index.mjs)
notify
1 COLLECTOR
mounts, bytes,
inodes, top dirs
2 GROWTH ANALYSER
24 h least-squares
days-to-full
3 ALERT ENGINE
thresholds + state
FIRING / RESOLVED
4 REPORTER
daily .txt
on-demand API
ALERT CHANNELS
log file · syslog
webhook · command
data/ & reports/ (plain files)
latest.json · history.jsonl (30 d) · topdirs.json · alert_state.json · alerts.jsonl/.log
HTTP SERVER (read-only GET)
/api/latest · /history · /alerts · /report · /team + dist/
polls every 5 s
REACT + TAILWIND + SHADCN DASHBOARD
Landing (crowd canvas) · Dashboard · Alerts
Report · FAQ + Team tabs (GitHub)

The main Logic for Our Project

Step	What the code does	Equivalent shell command
1. Find mounts	Reads /proc/mounts, skips pseudo filesystems like proc, sysfs and tmpfs, and removes duplicates	grep -vE 'proc|sysfs|tmpfs|cgroup' /proc/mounts
2. Disk usage	statfs() per mount: used = (blocks − bfree) × bsize, use% = used / (used + avail)	df -P -B1
3. Inodes	inodes_used = files − ffree, inode% = used / total	df -iP
4. Largest directories	Runs nice -n 19 du -x -d 1 -k <mount>, sorts by size, keeps the top 8	du -x -d1 -k /mnt | sort -rn | head
5. Growth	Appends a sample to history.jsonl each cycle, fits a least-squares line over the last 24 hours, and computes days_to_full = avail /	Compare today's df value with a stored one and divide
6. Alerts	Compares each value with the thresholds, builds a mount:kind key, and reports only changes	if ["$pct" -ge 90]; then logger -p user.crit "..."; fi
7. Report and exit code	Writes a text report; exit code is 0, 1 or 2	exit 2 for critical

Alerting and Reporting

Turning data into action

1. Warning threshold: 80% disk or inode usage.
2. Critical threshold: 90% disk or inode usage.
3. Growth warning: estimated full within 14 days.
4. Growth critical: estimated full within 3 days.
5. Sends alerts through logs, syslog, webhooks, or configured commands.
6. Generates daily reports for monitoring and auditing.

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Black and White cats

Home | Dashboard | Alerts | Report | FAQ

all clear

Dashboard

Shashanks-Laptop.local · last scan 9:55:57 PM · refreshes every 5s, samples every 30s · uptime 2d 3h

Total capacity	Used	Available	Active alerts
4 filesystems	24.8% of usable	free to non-root users	0 critical · 0 warning
470.92 GiB	116.93 GiB	353.99 GiB	0

/System/Volumes/iSCPreboot

/dev/disk1s1 · apfs

3.5% USED

Total	500.00 MiB
Used	17.39 MiB
Free	482.61 MiB
Status	OK

Inodes 36 / 49,41,996 (0.00%)

Growth / day	Full in
+0 B	not growing

Largest directories

/System/Volumes/iSCPreboot/SFR	5.48 MiB
/System/Volumes/iSCPreboot/SystemRecove…	44.00 KiB
/System/Volumes/iSCPreboot/959AA465-741…	16.00 KiB
/System/Volumes/iSCPreboot/5F9234CF-C7F4…	8.00 KiB
/System/Volumes/iSCPreboot/3D3287DE-280D…	4.00 KiB

/System/Volumes/Data

/dev/disk3s5 · apfs

24.5% USED

Total	460.43 GiB
Used	112.85 GiB
Free	347.58 GiB
Status	OK

Inodes 10,81,666 / 3,64,57,66,106 (0.03%)

Growth / day	Full in
+157.87 MiB	2254.5 days

Largest directories

/System/Volumes/Data/Users	45.04 GiB
/System/Volumes/Data/Applications	11.54 GiB
/System/Volumes/Data/private	7.27 GiB
/System/Volumes/Data/opt	6.16 GiB
/System/Volumes/Data/System	5.37 GiB

/Volumes/LVMData

/dev/disk2s3 · apfs

40.6% USED

Total	5.00 GiB
Used	2.03 GiB
Free	2.97 GiB
Status	OK

Inodes 110 / 3,11,31,510 (0.00%)

Growth / day	Full in
-60.02 KiB	not growing

Largest directories

No sub-directories reported (empty or unreadable).

/Volumes/LVMData 1

/dev/disk2s4 · apfs

Total	5.00 GiB
Used	2.03 GiB

Usage trend

Recorded samples from the collector

/System/Volumes/iSCPreboot | 1h | 6h | 1d | 7d

Black and White cats

Home | Dashboard | Alerts | Report | FAQ

all clear

Report

Download .txt

Storage report

Same text the collector writes to reports/ each day

```
================================================================================
 STORAGE REPORT  Shashanks-Laptop.local   13/10/2026, 9:56:27 pm
 darwin arm64  Darwin 25.6.0   uptime 2.16 days
================================================================================

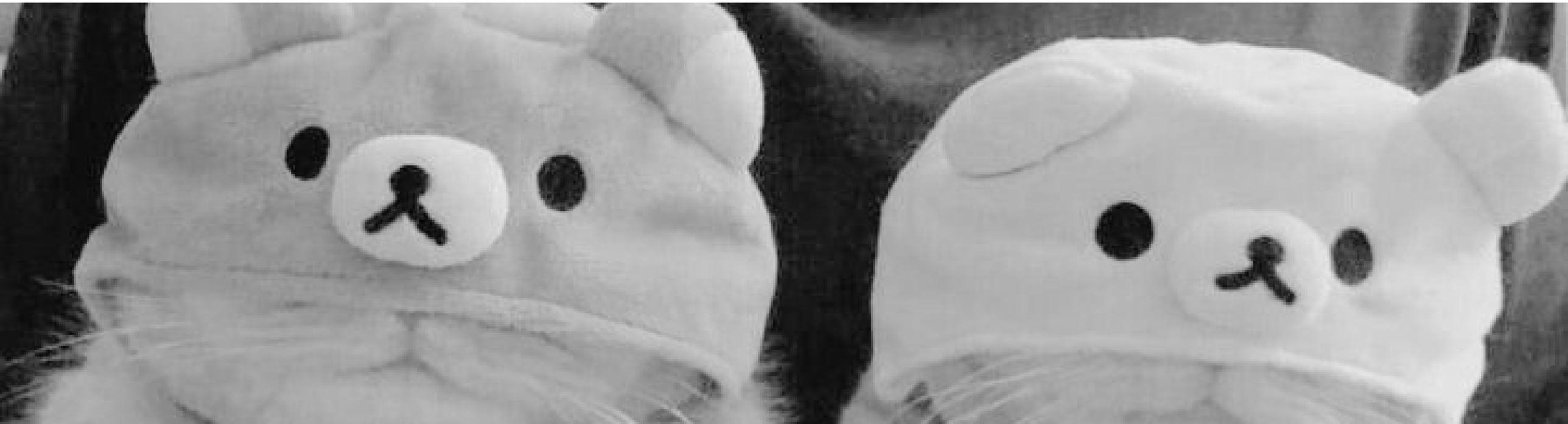
MOUNT                TYPE           SIZE       USED       FREE   USE%  INODE%  STATUS
--------------------------------------------------------------------------------
/System/Volumes/iSCPr apfs     500.00 MiB  17.39 MiB 482.61 MiB    3.5     0.0  OK
/System/Volumes/Data  apfs     460.43 GiB 112.85 GiB 347.58 GiB   24.5     0.0  OK
/Volumes/LVMData      apfs       5.00 GiB   2.03 GiB   2.97 GiB   40.6     0.0  OK
/Volumes/LVMData 1    apfs       5.00 GiB   2.03 GiB   2.97 GiB   40.6     0.0  OK

GROWTH (least-squares over last 24h of samples)
--------------------------------------------------------------------------------
/System/Volumes/iSCPreboot 0 B/day, 0 inodes/day, not growing
/System/Volumes/Data   52.43 MiB/day, 66408 inodes/day, full in 6788.0 days
/Volumes/LVMData       -58.21 KiB/day, -11 inodes/day, not growing
/Volumes/LVMData 1     -58.21 KiB/day, 0 inodes/day, not growing

LARGEST DIRECTORIES
--------------------------------------------------------------------------------
/System/Volumes/iSCPreboot:
      5.48 MiB  /System/Volumes/iSCPreboot/SFR
     44.00 KiB  /System/Volumes/iSCPreboot/SystemRecovery
     16.00 KiB  /System/Volumes/iSCPreboot/959AA465-741C-4F47-8A14-BEB4400FCBAE
      8.00 KiB  /System/Volumes/iSCPreboot/5F9234CF-C7F4-496F-B5F4-646CA2537157
      4.00 KiB  /System/Volumes/iSCPreboot/3D3287DE-280D-4619-AAAB-D97469CA9C71
/System/Volumes/Data:
     45.04 GiB  /System/Volumes/Data/Users
     11.54 GiB  /System/Volumes/Data/Applications
      7.27 GiB  /System/Volumes/Data/private
      6.16 GiB  /System/Volumes/Data/opt
      5.37 GiB  /System/Volumes/Data/System
/Volumes/LVMData:
/Volumes/LVMData 1:

ACTIVE ALERTS
```

TESTING AND VALIDATION



Applications and Future Scope

Applications

- Linux servers and virtual machines.
- Development environments and storage servers.
- Backup systems and data-processing machines.
- System administration labs.

Future enhancements

- Email and mobile notifications.
- Authentication and role-based access.
- Multi-server monitoring from one dashboard.
- Improved forecasting and anomaly detection.
- Configurable retention and historical trend visualizations.

THANK YOU